

Department of Computer Science

EXIT Exam Tutorial-2025

Data Communication & Computer Networks

Subnetting

The need for sub netting

- ▶ Classes of IP addresses offer a range from 256 to 16.8 million hosts.
- ▶ **Subnetting** separates a network into multiple logically defined segments, or subnets.
- ▶ To efficiently manage a limited supply of IP addresses, all classes can be subdivided into smaller sub networks or **subnets**.
- ▶ This process is known as **subnetting**.

The sub - netting process

- ▶ To create the sub network structure, **host** bits must be reassigned as **network** bits which is often referred to as **borrowing** bits.
- ▶ The **starting point** for this process is always the **leftmost** bit of the host.
 - ▶ the one closest to the last network octet.
- ▶ Subnet addresses include :
 - ▶ The Class A, Class B, and Class C network portion,
 - ▶ a **subnet field** and
 - ▶ a **host field**.

- ▶ The **subnet field** and the **host field** are created from the original host portion of the major IP address.
- ▶ This is done by assigning bits from the **host portion** to the original **network portion** of the address.
- ▶ Subnets have sub network ID (**subnet ID**) just as networks have network IDs.
- ▶ **Subnet IDs** are found by replacing all host fields with 0s.

Sub netting Advantages

- ▶ Make a network more manageable
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- ▶ Enables the network administrator to reduce broadcast domains
 - ▶ Since access to other subnets is only available through the services of a router, the network administrator can configure the router in different ways to **ensure security**.
 - ▶ For example, the network administrator can prevent hosts in subnet X from accessing a certain resource in subnet Y (see router **access lists** for more information)

How many bits should be borrowed?

- ▶ To determine the number of bits to be used, the network designer needs to calculate how many **hosts** the largest sub network requires and the number of **sub networks** needed
- ▶ **Large number of subnets** means **fewer hosts** and a large number of hosts means fewer subnets
- ▶ Total number of subnets is **$2^{\text{bits borrowed}}$**
- ▶ Total number of hosts is **$2^{\text{remaining host bits}}$**
- ▶ Example if three bits are borrowed from a class C address, total number of subnets is 8 (2^3) and total number of hosts is 32 (2^5)

Positional value of bits

Bits borrowed	1	2	3	4	5	6	7	8
Value	128	64	32	16	8	4	2	1

- Value is the **position** value of the bits borrowed
- Remember the value of each bit.
- It will help you in your calculations.
- Example

Q What is the value of 01010110 in decimal?

A $0 + 64 + 0 + 16 + 0 + 4 + 2 + 0 = 86$

Usable subnets

- ▶ Among the available subnets, it is not advised to use the following two subnets:
 - ▶ The subnet with all 0's in the subnet field
 - ▶ The subnet with all 1's in the subnet field
- ▶ If subnet zero (all 0's in the subnet field) is used, it means that a **network** and a **subnet** have the same address
- ▶ If the last subnet (all 1's in the subnet field) is used, it means that the **network broadcast address** and a **subnet** have the same address

Usable subnets & Usable Hosts

- ▶ Hence usable subnets will be $2^{\text{bits borrowed}} - 2$
- ▶ Example if three bits are borrowed from a class C address, total number of *usable subnets* is 6 ($2^3 - 2$) and total number of *usable hosts* is 30 ($2^5 - 2$)

Subnet masks

- ▶ For the subnet address scheme to work, every machine on the network must know which part of the host address will be used as the subnet address
- ▶ This is accomplished by assigning a **subnet mask** to each machine.
- ▶ A **subnet mask is a 32-bit** value that allows the recipient of IP packets to distinguish the network portion of the IP address from the host portion of the IP address

Subnet masks

- ▶ A subnet mask is composed of 1s and 0s where:
 - ▶ The **1s** in the subnet mask represent the positions that refer to **the network** or subnet addresses
 - ▶ The **0s** in the subnet mask represent the positions that refer to the **host** address

Default subnet masks

- ▶ Not all networks need subnets, meaning they use the **default subnet mask**
- ▶ This is basically the same as saying that a network doesn't have a subnet address. Here is default subnet mask for Classes A, B, and C
- ▶ Class A - network.node.node.node Subnet mask: **255.0.0.0**
- ▶ Class B network.network.node.node Subnet mask: **255.255.0.0**
- ▶ Class C - network. network.network.node Subnet mask: **255.255.255.0**
- ▶ These default subnet masks show the minimum number of 1's you can have in a subnet mask for each class.

Specifying subnets

- ▶ Example if three bits are borrowed from a class C address, the subnet mask is **255.255.255.224**
- ▶ Subnets may also be represented, in a slash format.
- ▶ For example, **/24** indicates that the total bits that were used for the **network** and **sub network** portion is 24
- ▶ The subnet mask 255.255.255.224 in slash format is **/27**
- ▶ **(224=11100000)**

Subnet mask (contd.)

- ▶ Number of bits borrowed from a class C address, positional value of each bit and resulting mask (in number and slash format)

Slash format	/25	/26	/27	/28	/29	/30	N/A	N/A
Mask	128	192	224	240	248	252	254	255
Bits borrowed	1	2	3	4	5	6	7	8
Value	128	64	32	16	8	4	2	1

Sub netting Class C addresses - Example 1

- ▶ Let us subnet the network address **192.168.10.0** with a subnet mask **255.255.255.192** or in slash format /26
- ▶ (192 is 11000000)

Q How many usable subnets do we have?

A Since 192 is 2 bits on (**11**000000), the answer would be $2^2 - 2$
 $= 2$

Q How many usable hosts per subnet do we have?

A We have 6 host bits off (**11****000000**), so the answer would be
 $2^6 - 2 = 62$ hosts

Sub netting Class C addresses - Example 1

Q What are the subnet IDs?

A We vary the borrowed bits (00, 01, 10, 11).

So the subnets are 192.168.10.0, 192.168.10.64, 192.168.10.128,
192.168.10.192

Q What are the valid or usable subnets

A The ones which do not have all 0's or all 1's in the subnet field,
namely 192.168.10.64 and 192.168.10.128

Sub netting Class C addresses - Example 1

Q What's the broadcast address for the valid subnets?

A The valid subnets start with 01 and 10. The broadcast address for these two addresses will have 01111111 and 10111111.

Which are 127 and 191. So the broadcast addresses will be

192.168.10.127 and 192.168.10.191. As a shortcut you can

follow this rule: The number right before the value of the next subnet is all host bits turned on and equals the broadcast address

Sub netting Class C addresses - Example 1

Q What are the valid hosts?

A These are the numbers between the subnet ID and broadcast address

The hosts for the first valid subnet are:

➤ 192.168.10.65, 192.168.10.66, ..., 192.168.10.126

The hosts for the second valid subnet are:

➤ 192.168.10. 129, 192.168.10.130, ..., 192.168.10.190

Class Exercises 1

- ▶ Given a network address 192.168.10.0/27, this time with or in slash format /27
- ▶ Determine the a subnet mask
- ▶ Find the number of subnet and usable subnets
- ▶ Find the number of host and usable hosts
- ▶ Find the block size
- ▶ Identify the valid subnet, Broadcast and Host Address



Sub netting Class C addresses - Example 2

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- ▶ Now let us subnet the network address 192.168.10.0, this time with a subnet mask 255.255.255.224 or in slash format /27

Q How many subnets do we have?

A Since 224 is 3 bits on (**111**00000), the answer would be
 $2^3 - 2 = 6$

Q How many hosts per subnet do we have?

A We have 6 host bits off (111**00000**), so the answer would be
 $2^5 - 2 = 30$ hosts

Q What are the subnet IDs?

A We vary the borrowed bits (000, 001, 010, 011, 100, 101, 110, 111). So the subnets are 192.168.10.0, 192.168.10.32, 192.168.10.64, 192.168.10.96, 192.168.10.128, 192.168.10.160, 192.168.10.192, 192.168.10.224

Q What are the valid or usable subnets

A 192.168.10.32, 192.168.10.64, 192.168.10.96, 192.168.10.128, 192.168.10.160, 192.168.10.192

Q What's the broadcast address for the valid subnets?

A The number right before the value of the next subnet is all host bits turned on and equals the broadcast address – 192.168.10.63, 192.168.10.95, 192.168.10.127, 192.168.10.159, 192.168.10.191, 192.168.10.223

Q What are the valid hosts?

192.168.10.33 – 192.168.10.62	192.168.10.129 – 192.168.10.161
192.168.10.65 – 192.168.10.94	192.168.10.161 – 192.168.10.193
192.168.10.97 – 192.168.10.128	192.168.10.193 – 192.168.10.222

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- ▶ Subnet the network address 192.168.10.0, with a subnet mask 255.255.255.248 (/28)

Q How many subnets do we have?

A Since 248 is 4 bits on (**1111**0000), $2^4 - 2 = 14$

Q How many hosts per subnet do we have?

A We have 6 host bits off (1111**0000**), $2^6 - 2 = 62$

Q What are the subnet IDs?

A We vary the borrowed bits (0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111). So the subnets ID's are :

192.168.10.0, 192.168.10.16, 192.168.10.32, 192.168.10.48,
192.168.10.64, 192.168.10.80, 192.168.10.96, 192.168.10.112,
192.168.10.128, 192.168.10.144, 192.168.10.160,
192.168.10.176, 192.168.10.192, 192.168.10.208,
192.168.10.224, **192.168.10.240**

Q What are the valid or usable subnets?

Q What's the broadcast address for the valid subnets?

A 192.168.10.31, 192.168.10.47, 192.168.10.63, 192.168.10.79,
192.168.10.95, 192.168.10.111, 192.168.10.127,
192.168.10.143, 192.168.10.159, 192.168.10.175,
192.168.10.191, 192.168.10.107, 192.168.10.223,
192.168.10.239

Sub netting Class C addresses - Example 3

Q What are the valid hosts?

192.168.10.17 – 192.168.10.30, 192.168.10.33 – 192.168.10.46
192.168.10.49 – 192.168.10.62, 192.168.10.65 – 192.168.10.78
192.168.10.81 – 192.168.10.94, 192.168.10.97 – 192.168.10.110
192.168.10.113 – 192.168.10.126, 192.168.10.129 – 192.168.10.142
192.168.10.145 – 192.168.10.158, 192.168.10.161 – 192.168.10.174
192.168.10.177 – 192.168.10.190, 192.168.10.193 – 192.168.10.206
192.168.10.209 – 192.168.10.222, 192.168.10.225 – 192.168.10.238

Example 3 (contd.)

Subnetwork #	Subnetwork ID	Host Range	Broadcast ID
0	192.168.10.0	.1--.30	192.168.10.31
1	192.168.10.32	.33--.62	192.168.10.63
2	192.168.10.64	.65--.94	192.168.10.95
3	192.168.10.96	.97--.126	192.168.10.127
4	192.168.10.128	.129--.158	192.168.10.159
5	192.168.10.160	.161--.190	192.168.10.191
6	192.168.10.192	.193--.222	192.168.10.223
7	192.168.10.224	.225--.254	192.168.10.255

Calculating class A and B networks

- ▶ The Class A and B sub netting procedure is identical to the process for Class C, except there may be significantly more bits involved
- ▶ Assigning 12 bits of a Class B address to the subnet field creates a subnet mask of 255.255.255.240 or /28.
- ▶ All eight bits were assigned in the third octet resulting in 255, the total value of all eight bits. Four bits were assigned in the fourth octet resulting in 240

Possible Class B subnet masks

255.255.128.0 (/17)

255.255.255.0 (/24)

255.255.192.0 (/18)

255.255.255.128 (/25)

255.255.224.0 (/19)

255.255.255.192 (/26)

255.255.240.0 (/20)

255.255.255.224 (/27)

255.255.248.0 (/21)

255.255.255.240 (/28)

255.255.252.0 (/22)

255.255.255.248 (/29)

255.255.254.0 (/23)

255.255.255.252 (/30)

Sub netting Class B addresses – Example 1

- ▶ 172.16.0.0 = Network address
- ▶ 255.255.192.0 = Subnet mask
- ▶ **Q** How many Subnets?
- ▶ **A** $2^2 - 2 = 2$.
- ▶ **Q** How many Hosts per subnet?
- ▶ $2^{14} - 2 = 16,382$. (6 bits in the third octet, and 8 in the fourth)
- ▶ **Q** Subnet IDs of valid subnets?
- ▶ **A** 172.16.64.0 and 172.16.128.0

Sub netting Class B addresses – Example 1

- ▶ **Q** Broadcast address for each subnet and valid hosts?
- ▶ **A** Below is the two subnets available and the address of each:
- ▶

Subnet	172.16.64.0	172.16.128.0
First host	172.16.64.1	172.16.128.1
Last host	172.16.127.254	172.16.191.254
Broadcast	172.16.127.255	172.16.191.255

Sub netting Class B addresses – Example 2

172.16.0.0 = Network address

255.255.240.0 = Subnet mask

Q How many Subnets?

A $2^4 - 2 = 14$

Q How many Hosts per subnet?

$2^{12} - 2 = 4094$

Q Subnet IDs of valid subnets?

A 172.16.16.0 and 172.16.32.0, ..., 172.16.224.0

Sub netting Class B addresses – Example 2

Q Broadcast address for each subnet and valid hosts?

A Below is the subnets available and the address of each:

Subnet	172.16.16.0	172.16.32.0	...
First host	172.16.16.1	172.16.32.1	...
Last host	172.16.31.254	172.16.47.254	...
Broadcast	172.16.31.255	172.16.47.255	...

Possible Class A subnet masks

255.128.0.0 (/9)	255.255.240.0 (/20)
255.192.0.0 (/10)	255.255.248.0 (/21)
255.224.0.0 (/11)	255.255.252.0 (/22)
255.240.0.0 (/12)	255.255.254.0 (/23)
255.248.0.0 (/13)	255.255.255.0 (/24)
255.252.0.0 (/14)	255.255.255.128 (/25)
255.254.0.0 (/15)	255.255.255.192 (/26)

Possible Class A subnet masks

255.255.0.0 (/16)

255.255.255.224 (/27)

255.255.128.0 (/17)

255.255.255.240 (/28)

255.255.192.0 (/18)

255.255.255.248 (/29)

255.255.224.0 (/19)

255.255.255.252 (/30)

Sub netting Class A addresses – Example 1

10.0.0.0 = Network address

255.255.0.0 (/16) = Subnet mask

Q Subnets?

A $2^8 - 2 = 254$

Q Hosts?

A $2^{16} - 2 = 65,534$

Q Valid subnets?

A 10.1.0.0, 10.2.0.0, 10.3.0.0, ..., 10.254.0.0

Sub netting Class A addresses – Example 1

Q Broadcast address for each subnet and valid hosts?

Subnet 10.1.0.0 ... 10.254.0.0

First host 10.1.0.1 ... 10.254.0.1

Last host 10.1.255.254 ... 10.254.255.254

Broadcast 10.1.255.255 ... 10.254.255.255

Sub netting Class A addresses – Example 2

10.0.0.0 = Network address

255.255.240.0 (/20) = Subnet mask

Q Subnets?

$$2^{12} - 2 = 4094$$

Q Hosts?

A $2^{12} - 2 = 4094$

.

Sub netting Class A addresses – Example 2

Q Valid subnets?

A

Subnet 10.1.0.0, 10.1.16.0,..., 10.255.224.0

First host 10.1.0.1, 10.1.16.1,..., 10.255.224.1

Last host 10.1.15.254, 10.1.31.254,..., 10.255.239.254

Broadcast 10.1.15.255, 10.1.31.255,..., 10.255.239.255

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ANDing

- ▶ The subnet mask gives routers the information required to determine in which network and subnet a particular host resides
- ▶ Routers make an AND operation between the subnet mask and the destination address (ANDing) to determine the subnet ID of the destination address. This information is required for routing purposes

Packet address	201.10.11.65	11001001.00001010.00001011.01000001
AND		
Mask	255.255.255.224	11111111.11111111.11111111.11100000
Subnetwork ID	201.10.11.64	11001001.00001010.00001011.01000000

ANDing (contd.)

- ▶ Another Example

- ▶ Address of host X is 192.168.54.84
- ▶ Subnet mask of host X is 255.255.255.224
- ▶ The sub network ID for host X: 192.168.54.84 AND 255.255.255.224 = **192.168.54.64** is the network address for host X